



**PRINCIPLES OF  
ENGINEERING  
DESIGN**

BSc. Mechatronic Engineering  
Third Year, Semester I

*Jomo Kenyatta University of  
Agriculture and Technology*

# RAINWATER HARVESTING

## *Pre-Filtration System*

---

From Concept to Final Design  
Decision-Making Process using Engineering Design Principles

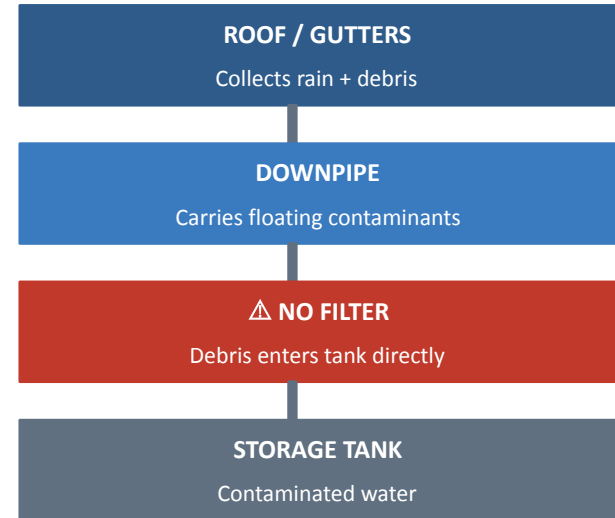
## 02 — Problem Statement

*Why pre-filtration matters for rainwater harvesting*

### The Challenge

- Roof surfaces collect large debris: leaves, dust, bird droppings, insects and micro-particles
- Heavy debris floats or suspends – conveyed directly into storage tanks
- First-flush diverters reduce initial contamination but floating debris still enters
- Piped networks not available in many rural areas – harvested rainwater is primary source
- Water bills, hard labour costs, and transport needs make clean harvesting critical

### Water Flow Path



**KEY INSIGHT:** Multi-stage pre-filtration removes large floating debris AND suspended fine particles — improving water quality before storage

## 03 — Objectives

*Design goals for the pre-filtration system*



### Minimize Debris Entry

Prevent leaves, dust, bird droppings and large particles from entering storage tanks



### Improve Water Quality

Reduce turbidity, germs and sediment; achieve up to 95% purity in rainy seasons



### Low Cost & Local

Design for local fabrication using affordable materials; accessible for rural households



### Easy Maintenance

Simple disassembly for cleaning gravel/mesh elements without specialist tools



### Modular & Compatible

Fits existing gutters and downpipes; scalable for rural and peri-urban installations



### Safe & Durable

Robust design for seasonal use with corrosion-resistant, pressure-rated materials

*Literature Review: Downpipe gravel filters achieve 90–95% water quality improvement; hybrid meshed+gravel approach is optimal for rural contexts (Wanduge et al.)*

# 04 — Design Approaches

*Three conceptual approaches developed and evaluated*

## Approach A

Complete / Meshed Gutter Sections

- Fine wire mesh gauze covers entire gutter length
- Meshed sections before AND after diverter
- Captures leaves and large debris at source
- Complex to install; leaves clog top over time
- Does not address suspended fine particles

*Simpler but limited filtration effectiveness*

## Approach B

Downpipe Gravel Filter

- Gravel bed integrated into downpipe section
- Screw-on design for different pipe sizes
- Threaded top cover – removable for cleaning/replacement
- Top and bottom covers are meshed
- Captures floating debris missed by first-flush diverter

*Good filtration; may be costly on large scale*

## Approach C

Hybrid: Meshed Sections + Downpipe Gravel

- Combines meshed gutter sections with downpipe gravel filter
- Coarse mesh removes large debris before diverter
- Gravel filter traps smaller particles after diverter
- Only fine dust and small leafy aggregates pass mesh
- Most captured by gravel; quality raised 5–15% extra

**Highest performance — SELECTED APPROACH**

→ *Evaluation via Decision Matrix* →

# 05 — Decision Matrix (Pugh's Method)

*Weighted evaluation of all design approaches*

## Weighted Criteria

10

**Safety / Water Quality**

Weight: 10/20

4

**Ease of Production**

Weight: 4/20

3

**Ease of Installation**

Weight: 3/20

2

**Durability**

Weight: 2/20

1

**Cost**

Weight: 1/20

| Approach            | S/Q (×10) | EP (×4) | EI (×3) | D (×2) | C (×1) | Total   | Score/20 |
|---------------------|-----------|---------|---------|--------|--------|---------|----------|
| A – Gutter Mesh     | 6(60)     | 0(0)    | 9(27)   | 10(20) | 5      | 40(152) | 7.6      |
| A – Meshed Sections | 7(70)     | 7(28)   | 10(30)  | 9(18)  | 8      | 41(154) | 7.7      |
| B – Gravel Filter   | 8(80)     | 6(24)   | 9(27)   | 10(20) | 8      | 40(159) | 7.95     |
| C – Hybrid ✓        | 9(90)     | 7(28)   | 9(27)   | 10(20) | 7      | 42(172) | 8.6 ★    |

✓ **SELECTED: Hybrid Approach C** — Highest overall score (8.6/10), best balance of water quality, practicality and cost

# 06 — Final Design: PED Filter Assembly

Modular 3-part uPVC screw-filter — hybrid implementation

|       |                                                                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MAT.  | <p><b>Unplasticized PVC (uPVC)</b></p> <p>Corrosion-resistant, pressure-rated thermoplastic. Well-suited to fluid filtration — no plasticizers means higher rigidity and chemical resistance.</p> |
| THD.  | <p><b>M100×6 Thread System</b></p> <p>External 6g (male) × Internal 6H (female). ISO 724 compliant medium-clearance fit. Hand-assembly possible without galling.</p>                              |
| FUNC. | <p>Inline downpipe filter; captures floating debris + sediment.</p> <p>600 mm body + 273 mm connectors × 2. Easy field clean.</p>                                                                 |

|                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Assembly Overview</b></p> <p>① <b>Filter Connector (×2)</b></p> <p>273 mm   Ø107.6 mm   M100×6 6g</p> <p>② <b>Locking Sleeve (×2)</b></p> <p>140 mm   Ø107.6 mm   M100×6 6H</p> <p>③ <b>Filter Main Body (×1)</b></p> <p>600 mm   Ø119.62 mm   M100×6 6g both ends</p> <p>Wall Thickness: <math>(\text{Ø}99.62 - \text{Ø}75.6) / 2 = 12.01 \text{ mm}</math>   All 5 parts: uPVC</p> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# 07 — Component Breakdown

Detailed dimensions — 4 drawing sheets

| PART 01                                                                         |                     |
|---------------------------------------------------------------------------------|---------------------|
| Filter Connector                                                                | Qty: 2 per assembly |
| Overall Length                                                                  | 273 mm              |
| Threaded Length                                                                 | 200 mm              |
| Outer Diameter                                                                  | Ø107.6 mm           |
| Inner Bore (ID)                                                                 | Ø95.6 mm            |
| Mid Diameter                                                                    | Ø101.6 mm           |
| Thread OD                                                                       | Ø99.62 mm           |
| Thread Spec                                                                     | M100×6 6g × 140 mm  |
| M100×6 6g male thread engages locking sleeve. Collar seats against sleeve face. |                     |

| PART 02                                                                          |                           |
|----------------------------------------------------------------------------------|---------------------------|
| Locking Sleeve                                                                   | Qty: 2 per assembly       |
| Overall Length                                                                   | 140 mm                    |
| Outer Diameter                                                                   | Ø107.6 mm                 |
| Thread Spec                                                                      | M100×6 6H                 |
| Thread Class                                                                     | 6H – Close tolerance      |
| Fit pairing                                                                      | 6g external / 6H internal |
| Assembly                                                                         | Hand-assembly, no galling |
| ISO internal thread class 6H paired with 6g external forms medium-clearance fit. |                           |

| PART 03                                                                     |                     |
|-----------------------------------------------------------------------------|---------------------|
| Filter Main Body                                                            | Qty: 1 per assembly |
| Overall Length                                                              | 600 mm              |
| Inner Body Length                                                           | 320 mm              |
| Threaded Length                                                             | 290 mm              |
| Outer Diameter                                                              | Ø119.62 mm          |
| OD at Thread Zones                                                          | Ø99.62 mm           |
| Inner Bore (ID)                                                             | Ø75.6 mm            |
| Thread Spec                                                                 | M100×6 6g × 70 mm   |
| Full body OD steps from Ø119.62 mm at flanges to Ø99.62 mm at thread zones. |                     |

## 08 — Assembly Sequence & Function

*How the three parts interconnect and seal*

01

### Thread Connectors

Thread Filter Connector onto each end of Filter Main Body — M100×6 6g engagement, 70 mm deep

02

### Slide Locking Sleeves

Slide Locking Sleeve over each Filter Connector BEFORE threading connector onto body

03

### Seat Connector

Once connector is fully seated, wind Locking Sleeve back along M100×6 6H internal thread

04

### Compression Lock

Ø107.6 mm collar of connector bears against face of sleeve — creates positive compression lock

### WATER FLOW PATH

Rooftop & Gutters

Meshed Gutter Sections

First-Flush Diverter

Downpipe Gravel Filter

Storage Tank

*Maintenance: Unscrew connector to access and clean gravel/mesh elements — no specialist tools required*

# 09 — Design Evaluation & Compliance

Assessment against design report framework

| Requirement             | Target / Spec            | Method                      | Status |
|-------------------------|--------------------------|-----------------------------|--------|
| Material compliance     | uPVC (unplasticized)     | Parts list inspection       | ✓ MET  |
| Thread compatibility    | M100×6 — 6g/6H pair      | Drawing specification       | ✓ MET  |
| Modular assembly        | 3-part screw system      | Assembly drawing analysis   | ✓ MET  |
| Connector sealing face  | Ø107.6 mm collar contact | Dimensional verification    | ✓ MET  |
| Body length (filter)    | 600 mm overall           | Dimension drawing (Sheet 2) | ✓ MET  |
| Wall thickness (body)   | ≈12 mm (Ø99.62–Ø75.6)    | Calculated from dims        | ✓ MET  |
| Connector thread length | 140 mm engagement        | Sheet 3 specification       | ✓ MET  |

12.01 mm

Wall Thickness  
(Ø99.62 – Ø75.6) ÷ 2

ISO 724

Thread Standard  
Both 6g & 6H classes conform

90–95%

Debris Removal  
Expected hybrid performance

5 Parts

Total Components  
All uPVC, locally fabricable

# 10 — Conclusions & Recommendations

## Design Summary

- 3-part modular uPVC filter: Main Body + 2 Connectors + 2 Locking Sleeves
- M100×6 threaded joints (6g/6H, ISO 724) — hand-assembled in field
- 600 mm main body with 12 mm wall thickness — pressure-rated

## Hybrid System Advantages

- Cleaner water in tanks — less debris, lower turbidity
- Reduced tank cleaning frequency (less frequent maintenance)
- 90–95% debris removal; up to 95% purity in rainy seasons
- Scalable for household, schools and rural community use

## Next Steps & Recommendations

1. Physical prototype pressure testing under simulated rainfall conditions
2. Seal validation at each threaded joint (M100×6 6g/6H engagement)
3. Fatigue analysis under cyclic loading; UV degradation study for uPVC
4. Post-filtration: boiling or UV sterilisation can raise purity to 99%+